

ISHEETA VERMA

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EDUCATION

Binghamton University, State University of New York

Master of Science in Computer Science | GPA: 3.4/4.00

May 2025

Relevant Coursework: Social Media Data Science Pipeline, Design & Analysis of Algorithms, Database Systems, Distributed Systems, Functional PL, Machine Learning

Jaypee Institute of Information Technology Noida, India

Bachelor of Technology in Electronics and Communication

May 2021

TECHNICAL SKILLS

Languages: Python, Java, C++, C, HTML/CSS, Javascript, Typescript, Bash, SQL

Data Stores: MySQL, SQL Server, MongoDB, HDFS, PostgreSQL, Cassandra, OracleDB, Azure Data Lakes, Azure SQL, AWS (S3, Redshift, EC2, RDS, Lambda), GCP (BigQuery, CloudSQL), DynamoDB

Infra and Tools: Docker, Kubernetes, CI/CD (GitHub Actions, GitLab CI), Git, Linux, Unix, Hadoop, MapReduce, Jira, Apache Spark, Databricks, Kafka, Airflow, Power BI, Pandas, Snowflake, Azure Data Factory, Terraform

Domain Knowledge: Data Analysis, Visualization, Microservices, Machine Learning, API Development, System Design, OOPs

Certifications: Microsoft AZ-900 Fundamentals

PROFESSIONAL EXPERIENCE

DXC Technology, Associate Professional Software Engineer

Feb 2021- Aug 2023

- Accelerated release cycles by 50% by integrating **PyTest, JUnit, and Apache Airflow** into automated QA workflows, implementing test-driven development (TDD) practices and reducing post-deployment defects by 20%
- Designed **Spark/Airflow ETL pipelines** to process 10TB+ monthly structured and unstructured data from AWS S3 to Redshift, ensuring data integrity, debugging failures in distributed clusters, and improving processing speed by 40%
- Engineered a **Kafka-Flink** pipeline to process multi-terabyte daily financial data with millisecond latency, applying **TCP/IP** and event-driven messaging for personalization, fraud detection, and insights
- Improved **API and database responsiveness** by 30% through optimized **MongoDB queries, Redis caching**, and schema/index tuning in SQL/PostgreSQL, reducing analytics dashboard load times by 25% under peak workloads
- Reduced incident response time by 40% by deploying scalable **Python/Docker microservices on AWS (EC2, RDS, S3)** and integrating monitoring with Grafana and CloudWatch to debug latency and availability issues
- Improved **microservices** reliability and throughput by 25% by implementing **gRPC/ Kafka** for scalable, low-latency communication
- Enhanced client integration by 35% by designing RESTful APIs (Flask/FastAPI, PostgreSQL) with 99.99% uptime, lowering partner support tickets by 30% and strengthening system usability
- Strengthened data security via **role-based access control** in AWS S3/Redshift, improving compliance and client satisfaction by 40%
- Boosted user engagement conversion rates by 12% by integrating **React + TypeScript** dashboards with **GraphQL/REST APIs** and validating application flows via **A/B testing**
- Increased reporting speed by 35% with Snowflake fact-dimension schemas using clustering, materialized views for enterprise analytics
- Reduced deployment time by 50% through **GitHub Actions**-based CI/CD automation in agile sprints, enabling fast, error-free releases

Elwave Digitech(I) Private Limited, Software Engineer

June 2020- Jan 2021

- Engineered** real-time backend services in **Java/Python** using SOLID principles and design patterns or scalable, fault-tolerant microservices for high-availability systems
- Developed **event-driven pipelines (TCP/IP + REST APIs)** to integrate on-prem transactional systems with cloud analytics and CRM platforms, ensuring secure, reliable data exchange
- Optimized **ETL workflows with multi-threaded Java/SQL**, boosting throughput and reducing runtime by 35% across large-scale reporting workloads

PROJECT EXPERIENCE

E-commerce Data Modernization with Snowflake

Dec 2024 – Jan 2025

- Designed and implemented a Snowflake data warehouse, integrating Azure Data Lake to streamline ingestion, cutting processing time from 24 to 1 hour, and enabling real-time analytics for 5M customers via a three-tier medallion architecture
- Leveraged Snowflake's **Streams, Time Travel, and Zero Copy Cloning** to enhance data integrity, automate data transformations, and support advanced analytics and machine learning applications

Medical Imaging Pipeline and Pneumonia Detection Using Deep Learning

Nov 2024 – Dec 2024

- Developed a pipeline for DICOM file preprocessing and visualization using **PyDicom, SimpleITK, and nibabel**, enhancing data quality and optimizing 2D/3D imaging workflows through spatial resampling and orientation standardization, reducing processing time by 30%
- Built a ResNet18 CNN for pneumonia detection with 90% accuracy using preprocessing, normalization, and Class Activation Maps

Sentiment and Engagement Analysis of Social Media Platforms

Oct 2024 – Nov 2024

- Built an analytics dashboard using **Flask, PostgreSQL, and Plotly** to visualize user engagement, sentiment analysis, and toxic content trends across online platforms
- Developed **real-time data pipelines** and advanced relational database queries to track platform-specific metrics, analyze political events' impact, and detect hate speech trends